UNIT – III

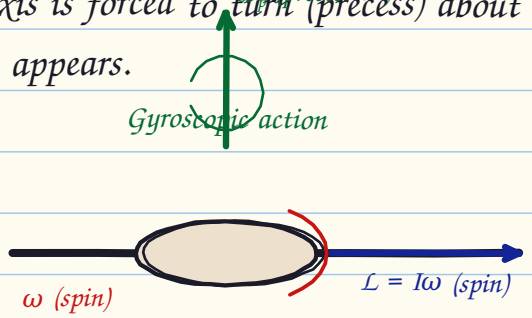
Gyroscope & Gear Profile

Subject : Theory of Machines (23A03403)

Syllabus: Principle of gyroscope, gyroscopic effect in aeroplane, ship, car & two-wheeler ; involute & cycloidal profiles, gear parameters, fundamental law of gearing & conjugate action, contact ratio, interference/undercutting, helical, bevel, worm, rack & pinion, epicyclic & regular gear trains.

1. Gyroscope – Principle

A gyroscope is a spinning rotor mounted so that its axis can turn (gimbals). When the spin axis is forced to turn (precess) about an axis perpendicular to it, a gyroscopic couple appears.

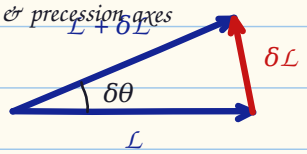


$\delta L = L\,\delta\theta = I\omega\,\delta\theta$

$dL/dt = I\omega\,\omega_p$

Active couple $C = I\omega\,\omega_p$

⊥ to both spin & precession axes



Reaction couple on frame = $-C$

- Spin velocity ω (rad/s) about the spin axis ; precession (angular velocity of the axis) ω_p about the perpendicular axis ; moment of inertia of rotor I.
- Angular momentum $L = I\omega$ (vector along spin axis by right-hand rule). In time δt the spin axis turns by $\delta\theta \Rightarrow$ change in momentum $\delta L = L\,\delta\theta$ (perpendicular to L).

$$C = dL/dt = I\,\omega\,\omega_p \text{ (active gyroscopic couple)}$$

- Direction : the active couple vector is $\omega_p \times L$ (right-hand rule) – the axis tends to turn so that spin vector moves towards the precession vector. The **reaction couple** on the supporting frame (bearings) is equal and opposite.

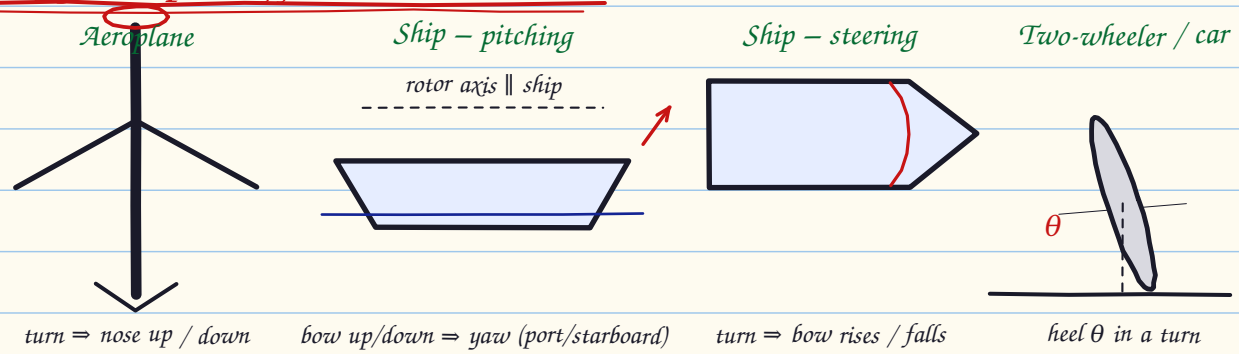
- Units : I in kg·m² = $m\,k^2$; ω_p in rad/s ; C in N·m. For harmonic precession (pitching of ship) $\omega_{p,max} = (2\pi/T)\,\varphi_{max}$ (φ_{max} = amplitude in rad, T = period).

Rules to find the reaction couple

- Draw the spin vector (right-hand screw). Draw the precession vector. Active couple = turn the spin vector towards the precession vector through 90° ; reaction couple = reverse.
- When the spin axis and precession axis are parallel (rolling of a ship), no gyroscopic couple. If they are inclined at angle θ : $C = I\omega\omega_p \sin\theta$.
- Applications : gyrocompass, gyro-stabiliser for ships, aircraft attitude indicators, inertial navigation, spinning top, bicycle stability, projectile (rifling), gyro-seismometer.

Ex 1. Ship rotor mass 3000 kg, k = 0.6 m, 1500 rpm ; ship pitches simple harmonically with amplitude 6° and period 20 s : I = 3000 × 0.36 = 1080 kg·m² ; ω = 157.1 rad/s ; $\omega_{p,max} = (2\pi/20)(6\pi/180) = 0.0329$ rad/s ⇒ C_{max} = 1080 × 157.1 × 0.0329 = **5580 N·m**.

2. Gyroscopic Effects on Vehicles



(a) Aeroplane

- Rotating parts (engine, propeller) have spin about the longitudinal axis. When the aeroplane **turns** (yaw precession, $\omega_p = v/R$ about the vertical) the reaction couple acts in the pitching plane (nose up/down) :

$$C = I\,\omega\,\omega_p = I\,\omega\,(v/R)$$

- Rule (propeller rotating clockwise viewed from the rear) : turn left ⇒ nose rises (tail falls) ; turn right ⇒ nose falls. For anticlockwise rotation the effects are reversed.

- Pitching of the aircraft (nose up/down) produces a couple about the vertical axis (tends to yaw it) ; rolling produces none (axes parallel).

> Ex 2. Engine + propeller $I = 36\ kg{\cdot}m^2$, 2400 rpm clockwise from rear ; aeroplane flying at 400 km/h turns right on a 600 m radius. $\omega = 251.3$ rad/s ; $v = 111.1$ m/s $\Rightarrow \omega_p = 0.1852$ rad/s ; $C = 36 \times 251.3 \times 0.1852 =$ 1676 N·m ; it tends to depress the nose and raise the tail (right turn, clockwise).

(b) Ship

- Rotor (turbine) axis along the ship's length. Three motions : **steering** (yaw), **pitching** (up-down oscillation of bow), **rolling** (no effect).
- Steering : same as aeroplane – with rotor clockwise looking from the stern, turning left raises the bow ; turning right lowers the bow (effect is a couple in the vertical plane).
- Pitching (bow moving up) ⇒ reaction couple yaws the ship : for a clockwise rotor viewed from the stern, upward pitching ⇒ ship tends to turn to starboard (right) and downward pitching ⇒ to port (left).
- Effect on bearings : reaction couple resisted by a pair of reactions $\pm C/L$ at the bearings (L = distance between bearings). For pitching of period T, maximum angular velocity occurs at the mean position.

(c) Two-wheeled vehicle taking a turn

- Wheels spin about horizontal axis ; during a turn at radius R the vehicle precesses at $\omega_p = v/R$. If the vehicle is inclined at θ to the vertical, the effective precession component is $\omega_p \cos\theta$ ⇒

$$C_g = (I_{w1}\omega_{w1} + I_{w2}\omega_{w2} \pm I_e\omega_e) (v/R) \cos\theta$$

- Engine rotor sign : '+' if it rotates in the same direction as wheels, '–' if opposite. The reaction gyroscopic couple tends to overturn the vehicle outwards (away from the centre).
- Centrifugal force $F_c = m v^2/R$ acts at height h of the CG : centrifugal couple = $(mv^2/R)\, h \cos\theta$ (also outward). The weight couple $mgh \sin\theta$ must balance them :

$$\tan\theta = v^2/(gR) + C_{g0}/(m\, g\, h)\ (C_{g0} = \text{gyro couple for } \theta = 0)$$

Ex 3. Motor cycle + rider 200 kg, $h = 0.6$ m, $v = 36$ km/h (10 m/s), $R = 30$ m. Wheel $I = 0.4$ kg·m² each, wheel radius 0.3 m ($\omega_w = 33.3$ rad/s) ; engine rotor $I_e = 0.2$ kg·m² at 5 × wheel speed, same direction. $C_{g0} = (2\times0.4\times33.3 + 0.2\times166.7)(10/30) = 20$ N·m ; $\tan\theta = 100/(9.81\times30) + 20/(200\times9.81\times0.6) = 0.340 + 0.017 = 0.357 \Rightarrow \theta = 19.6°$ to the vertical.

(d) Four-wheeled vehicle (car) taking a turn

- Three effects on the vertical reactions of the wheels : (i) weight mg/2 on each side ; (ii) **gyroscopic couple** from wheels & engine $C_g = (4I_w\omega_w \pm I_e\omega_e G)(v/R)$, producing $\Delta P = C_g/(2a)$ on outer wheels (+) and inner (–) where 2a = track width ; (iii) **centrifugal couple** $(mv^2/R)h \Rightarrow \Delta P = mv^2h/(2aR)$.

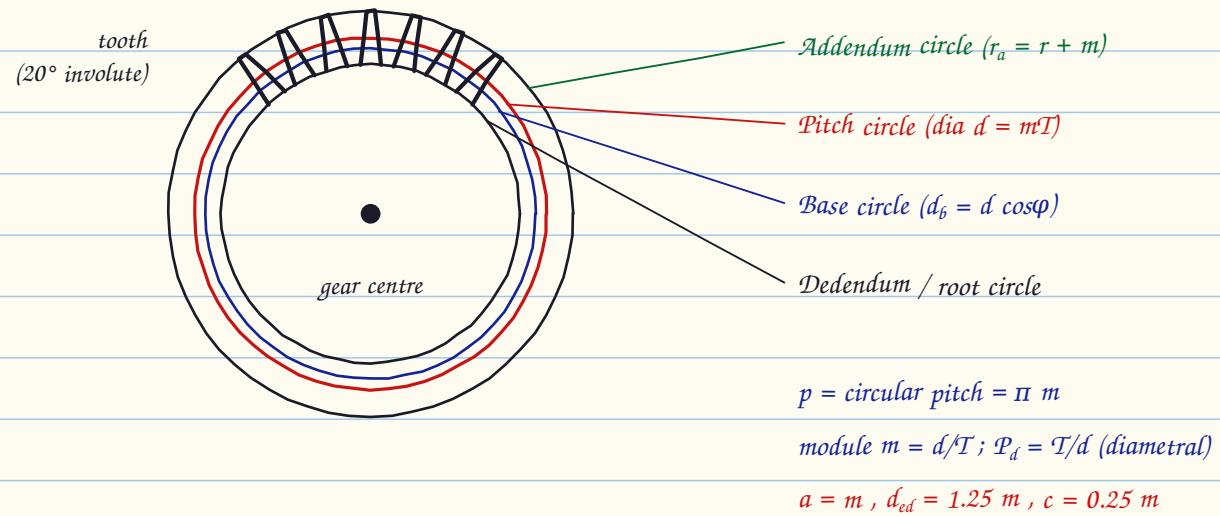
$$P_{outer} = mg/2 + (1/2a)[\, C_g + m\,v^2\,h\,/R\,] \qquad P_{inner} = mg/2 - (1/2a)[\, C_g + m\,v^2\,h/R\,]$$

- Limiting speed for overturning (inner wheels lift off, $P_{inner} = 0$) ; skidding when $F_c > \mu\, mg$ $(v^2/R > \mu g)$ – whichever occurs first.
- Stability remedy : lower CG, wider track, bank the road, opposite-rotating engine rotor (gyro compensation), tyres with high μ.

3. Gear Profile – Basics

Gears transmit motion and power between shafts with *positive drive, constant velocity ratio and compact size*. Gear types : spur (parallel shafts), helical, herringbone, bevel (intersecting), worm / hypoid / crossed-helical (skew), rack & pinion.

Spur gear terminology



Spur gear terminology

- **Pitch circle** : imaginary circle rolling without slip ; diameter d. **Module** $m = d/T$ (mm) – standard 1, 1.25, 1.5, 2, 2.5, 3, 4, 5, 6, 8, 10… ; **diametral pitch** $P_d = T/d$; **circular pitch** $p = \pi d/T = \pi m$; $p\,P_d = \pi$.
- **Addendum** $a = m$ (distance pitch → tip) ; **dedendum** $1.25\,m$; **clearance** $c = 0.25\,m$; **whole depth** $2.25\,m$; **working depth** $2m$. **Tooth thickness** = space width = $p/2$ (no backlash) ; backlash = space − thickness.
- **Pressure angle** φ : angle between common normal and common tangent at the pitch point ($14½°$, $20°$, $25°$). **Base circle** $d_b = d\cos\varphi$. **Base pitch** $p_b = p\cos\varphi$. **Face width** b (≈ 8–12 m). **Centre distance** $C = (d_1 + d_2)/2 = m(T_1 + T_2)/2$.
- **Velocity ratio** $VR = \omega_1/\omega_2 = N_1/N_2 = d_2/d_1 = T_2/T_1$ (external gears – opposite rotation).

Fundamental law of gearing (conjugate action)

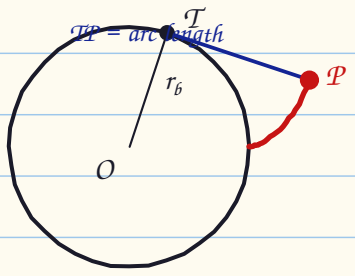
For a constant velocity ratio the common normal at the point of contact of the mating teeth must always pass through a fixed point P (the pitch point) on the line of centres :

$$\omega_1/\omega_2 = O_2P / O_1P = r_2/r_1$$

- Profiles that satisfy this law are called **conjugate profiles.** Commonly used : involute and cycloidal.

Involute profile

Generation of an involute



Properties

- TP (tangent) is the normal to the curve at P
- involute starts at the BASE circle ; none inside
- shape depends only on r_b ($= r\cos\varphi$)
- inv $\varphi = \tan\varphi - \varphi$; $r = r_b/\cos\varphi$
- centre distance change ⇒ ratio unchanged

- Involute = curve traced by the end of a string unwound from the base circle. Tooth profile from the base circle to the addendum circle. The common normal = tangent to base circles = the **line of action** – a straight line inclined at φ to the common tangent through P.

- Properties : (i) normal to the involute is tangent to the base circle ; (ii) the involute ratio depends only on base radii ⇒ a small change of centre distance does not affect the velocity ratio ; (iii) pressure angle constant during mesh (single line of action) ⇒ constant force direction and smooth running ; (iv) easy manufacture (straight-sided rack cutter, hobbing). Involute function $\text{inv}\,\varphi = \tan\varphi - \varphi$.

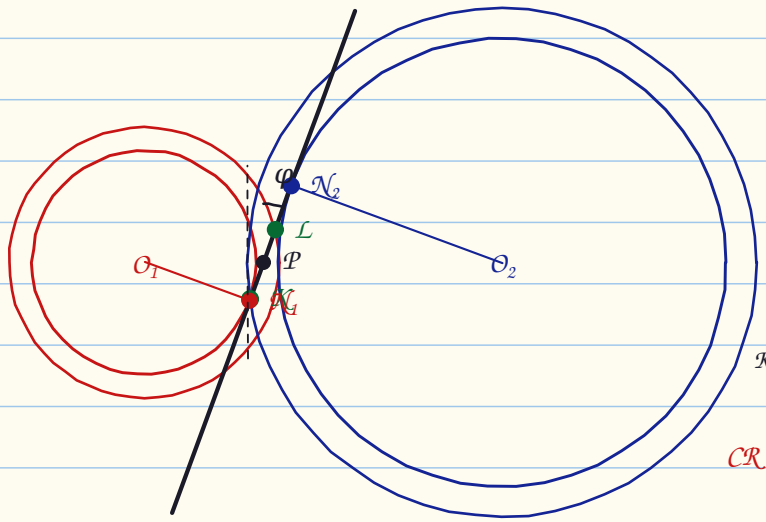
Cycloidal profile

- Face = **epicycloid** (rolling circle outside pitch circle) and flank = **hypocycloid** (rolling inside).

Item	Involute	Cycloidal
Profile	single curve face & flank	epicycloid + hypocycloid
Pressure angle	constant	varies (zero at pitch point)
Centre-distance change	ratio unchanged	ratio changes – exact distance needed
Interference	possible (small T)	none
Wear, strength	flank weaker at root	stronger, less wear (convex-concave)
Manufacture	easy, cheap	difficult
Use	almost all gears	clock & watch gears, heavy-load low-speed

4. Path of Contact, Arc of Contact & Contact Ratio

Path of contact $K L$, line of action $N_1 N_2$



KP : path of approach

PL : path of recess

$CR = KL / (\pi\, m \cos\varphi)$

- **Path of contact** KL : locus of the contact point – part of the line of action between the addendum circles. K = where the addendum circle of the **driven** gear cuts the line of action (start), L = where the addendum circle of the driver cuts it (end).
- With R_{a1}, R_{a2} = addendum radii and R_{b1}, R_{b2} = base radii, φ = pressure angle, C = centre distance :

$$KP = \sqrt{(R_{a2}^2 - R_{b2}^2)} - R_2 \sin\varphi \qquad PL = \sqrt{(R_{a1}^2 - R_{b1}^2)} - R_1 \sin\varphi$$

$$\text{Path of contact } KL = \sqrt{(R_{a1}^2 - R_{b1}^2)} + \sqrt{(R_{a2}^2 - R_{b2}^2)} - C \sin\varphi$$

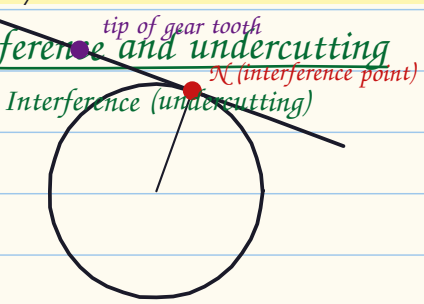
- **Arc of contact** = $KL / \cos\varphi$. **Contact ratio** (average number of teeth in contact) :

$$CR = \text{arc of contact} / \text{circular pitch} = KL / (\pi\, m \cos\varphi) = KL / p_b$$

- CR must be > 1 ($\geq$ 1.2 practically ; 1.4–1.8 for spur) so that a new pair engages before the previous leaves – otherwise noise & impact. A CR of 1.6 means 2 pairs in contact 60 % of time and 1 pair 40 %.

Ex 4. m = 5 mm, T_1 = 20, T_2 = 40, φ = 20°, addendum = m. R_1 = 50, R_2 = 100, R_{a1} = 55, R_{a2} = 105, R_{b1} = 46.98, R_{b2} = 93.97, C = 150 mm. $KP = \sqrt{(105^2 - 93.97^2)} - 100 \sin20° = 46.85 - 34.20 = 12.65$; $PL = \sqrt{(55^2 - 46.98^2)} - 50 \sin20° = 28.59 - 17.10 = 11.49$; KL = 24.14 mm ⇒ arc = 25.69 mm ; CR = 25.69/15.71 = 1.64.

Interference and undercutting



Occurs when the tip of the driven gear tooth cuts the pinion flank below the base circle

Avoid by : $T \geq T_{min} = 2A_w/\sin^2\varphi$ (rack)

- raise φ (20°, 25°)
- stub teeth
- profile shift
- more teeth on pinion
- reduce gear tip radius

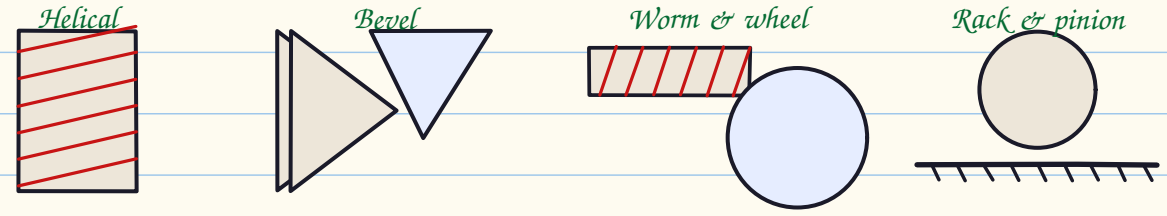
- **Interference** : tip of a tooth of one gear contacts the non-involute portion (below base circle) of the mating tooth, so the teeth jam/cut. It occurs when the addendum circle of the gear cuts the line of action beyond the tangent point N_1. **Undercutting** : the cutter removes the root of the pinion tooth ⇒ weak tooth.

- **Minimum number of teeth** on the pinion (gear ratio $G = T_G/T_P$, addendum A_w m) :

$$T_{min} = 2A_w / [\sqrt{(G^2 + (1 + 2G)\sin^2\varphi)} - G] \text{ (rack, } G \to \infty : T_{min} = 2A_w/\sin^2\varphi)$$

- For $\varphi = 20°$, $A_w = 1$: rack → 17.1 (18 practically 17), $G = 1$ → 12.3 ⇒ 13, $G = 2$ → 14.2 ⇒ 15. For 14½° → 32 teeth ; for 25° → 12 teeth.
- **Remedies** : increase pressure angle, stub teeth ($A_w < 1$), more teeth on pinion, profile shifting (addendum modification / X-gears), reduce gear addendum, adjust centre distance.

5. Types of Gears



Helical gears

- Teeth cut at a helix angle ψ ; contact starts at one end and spreads gradually – smoother, quieter, higher load capacity and contact ratio ; but produce axial thrust $F_a = F_t \tan\psi$ (use herringbone / double helical to cancel).
- Normal module $m_n = m\cos\psi$ (m = transverse module) ; pitch dia $d = T\, m_n/\cos\psi$; **virtual number of teeth** $T_v = T/\cos^3\psi$ (used for strength and interference). Shafts parallel (opposite hand) or crossed (same hand). Minimum teeth lower than spur.

> Ex 5. Helical : $T = 30$, $m_n = 4$ mm, $\psi = 25° \Rightarrow m = 4/0.9063 = 4.41$ mm ; $d = 132.4$ mm ; $T_v = 30/0.9063^3 = 40.3$.

Bevel gears

- Intersecting shafts (usually 90°). Pitch cones roll without slip ; pitch cone angles δ_1, δ_2 with $\tan\delta_1 = T_1/T_2$ ($\Sigma = 90°$) ; types : straight, spiral, zerol, hypoid. Virtual (Tredgold) teeth $T_v = T/\cos\delta$.

> Ex 6. $T_1 = 20$, $T_2 = 40$, shafts 90° : $\delta_1 = \tan^{-1}(0.5) = 26.6°$, $\delta_2 = 63.4°$; $T_{v1} = 20/0.894 = 22.4$; $T_{v2} = 40/0.447 = 89.4$.

Worm and worm wheel

- Worm (screw) with Z starts drives the wheel with T_G teeth ; shafts skew (usually 90°) ; velocity ratio = T_G/Z (single reduction up to 100 : 1). Lead $L = Z \times$ axial pitch ; lead angle $\tan\lambda = L/(\pi\, d_w)$.
- Efficiency $\eta = (\cos\varphi_n - \mu\, \tan\lambda)/(\cos\varphi_n + \mu\, \cot\lambda)$; **self-locking** when $\lambda <$ friction angle ($\eta < 50\ \%$) – used in hoists, lifts, steering gear. Sliding action ⇒ heat, bronze wheel, oil.

Rack & pinion

- Rack = gear of infinite radius (straight pitch line, straight sides) : $v = \omega\, r$, linear travel = $\pi\, d$ (per rev) ; used in lathe carriage, steering, rack railway, jacks, drilling machines.

6. Gear Trains

Simple, compound and reverted trains

- **Simple** : one gear per shaft ; idler changes direction only. Speed ratio = T_{last}/T_{first} (idlers cancel). **Compound** : two or more gears on one shaft :

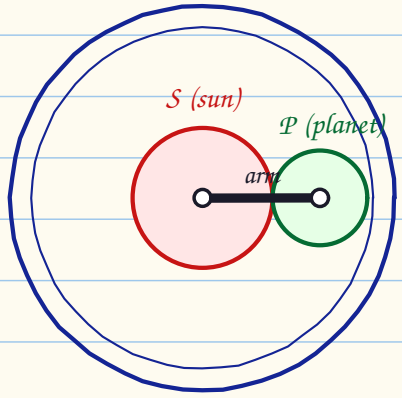
$$N_{last}/N_{first} = (T_1T_3\ldots)/(T_2T_4\ldots)\ \text{(product of drivers / product of followers)}$$

- **Reverted** : first and last shafts coaxial – centre distances equal : $m(T_1 + T_2) = m(T_3 + T_4)$ (lathe back gear, clocks).

Ex 7. $A(20)$ drives $B(50)$; $C(15)$ on B's shaft drives $D(60)$: speed ratio $N_A/N_D = (50/20)(60/15) = 10$; if $N_A = 1000$ rpm ⇒ $N_D = 100$ rpm (same direction – two meshes).

Epicyclic (planetary) gear trains

Epicyclic (planetary) gear train



R (ring / annulus) $T_R = T_S + 2T_P$

Tabular method (arm A, sun S, planet P, ring R)

Condition	Arm	Sun	Planet	Ring
1 Arm fixed, S turns +1	0	+1	$-T_S/T_P$	$-T_S/T_R$
2 Add +y to all	y	y	y	y
Total	y	y+1	$y - T_S/T_P$	$y - T_S/T_R$

Ex: ring fixed ⇒ $y = T_S/T_R$ ... see notes

- Axes of one or more gears move about the axis of the sun gear – carried by the **arm.** Compact, high ratio in small space, two DOF (one input fixed ⇒ 1 DOF). Uses : automatic transmission, differential, hoists, speed reducers, clocks.
- **Tabular method** : (1) arm fixed, give sun +1 turn ; find relative turns of other gears (external mesh reverses sign, internal mesh keeps sign) ; (2) add +y turns to all (arm turn y) ; (3) apply given conditions (a gear fixed or known speed) and solve for y.

Ex 8. Sun S(20), planet P(20), ring R(60 = 20 + 2×20). Ring fixed, sun input 1000 rpm : row 1 : arm 0, S +1, P −1, R −1/3 ; row 2 (add y) : arm y, S y + 1, P y − 1, R y − 1/3. R fixed ⇒ y = 1/3 ⇒ S = 4/3. Ratio S/arm = 4 ⇒ arm speed = 1000/4 = 250 rpm (same direction) ; planet = (1/3 − 1) × 750 = −500 rpm.

Ex 9. Simple epicyclic : gear A (40 teeth) fixed, gear B (30 teeth) meshing and carried by the arm, arm 100 rpm : B = arm $(1 + T_A/T_B)$ = 100(1 + 40/30) = 233 rpm in the same direction.

- **Differential** of an automobile : bevel epicyclic ; on a straight road both axles turn at the cage speed ; in a turn $N_L + N_R = 2\,N_{cage}$.
- Torque : the sum of the external torques on the train is zero ⇒ $T_{in} + T_{out} + T_{fixed} = 0$; power loss small ⇒ $T_{in}\,\omega_{in} \approx T_{out}\,\omega_{out}$.

7. Additional Solved Problems

P1 (gyro car). A car wheel (I_w = 1.0 kg·m², rolling radius 0.35 m) at 72 km/h (20 m/s) rounds a 50 m curve : ω_w = 57.1 rad/s ; ω_p = 0.4 rad/s ; C_{wheel} = 1.0 × 57.1 × 0.4 = 22.9 N·m per wheel ; four wheels 91.4 N·m (equal in direction, effect shared on outer/inner wheels).

P2 (gear pair). m = 4 mm, T_1 = 24, T_2 = 60 ⇒ d_1 = 96, d_2 = 240, C = 168 mm ; $p = \pi m$ = 12.57 mm ; VR = 2.5 ; $d_{b1} = 96 \cos 20° = 90.2$ mm.

P3 (minimum teeth). Pinion & gear equal (G = 1), φ = 20°, full depth : $T_{min} = 2/(\sqrt{(1 + 3 \sin^2 20°)} - 1) = 2/0.1623 = 12.3 \Rightarrow$ 13 teeth.

P4 (worm). Single-start worm, T_G = 40 ⇒ ratio 40 : 1. If worm turns 1200 rpm ⇒ wheel 30 rpm. Lead angle for d_w = 50 mm, axial pitch 12.57 mm : $\tan\lambda = 12.57/(\pi \times 50) = 0.080 \Rightarrow \lambda = 4.6°$ (self-locking if $\mu > 0.08$).

P5 (contact ratio). Path of contact KL = 20 mm, m = 4 mm, φ = 20° : $p_b = \pi m \cos\varphi$ = 11.81 mm ⇒ CR = 20/11.81 = 1.69 (about 2 pairs in contact for 69 % of the time).

Short answers – Quick Revision

- **Why does a ship's rolling cause no gyroscopic effect?** The roll axis is parallel to the rotor spin axis – no precession perpendicular to the spin vector.
- **Why is involute preferred?** Constant pressure angle and velocity ratio insensitive to centre-distance errors ; easy to cut.
- **Why do helical gears run quieter?** Gradual engagement along the face ; higher contact ratio.
- **How to increase the contact ratio?** More teeth, lower pressure angle, longer addenda.
- **When is an epicyclic train preferred?** High reduction/compact coaxial layout, and variable speed ratio by holding different members.

8. Quick Formula Sheet – Unit III

Item	Formula
Gyroscopic couple	$C = I\omega\omega_p$; $\omega_p = v/R$
Aeroplane / ship	$C = I\,\omega\, v/R$; ship pitching $\omega_{p,max} = (2\pi/T)\varphi_{max}$
Two-wheeler heel	$\tan\theta = v^2/gR + C_{g0}/(mgh)$
Gear	$m = d/T$; $p = \pi m$; $d_b = d\cos\varphi$; $C = m(T_1+T_2)/2$

Item	Formula
Fundamental law	$\omega_1/\omega_2 = O_2P/O_1P$
Path of contact	$KL = \sqrt{(R_{a1}^2 - R_{b1}^2)} + \sqrt{(R_{a2}^2 - R_{b2}^2)} - C\sin\varphi$
Contact ratio	$CR = KL/(\pi\, m\cos\varphi)$
Min. teeth	$T_{min} = 2A_w/(\sqrt{(G^2+(1+2G)\sin^2\varphi)} - G)$
Helical	$T_v = T/\cos^3\psi$; $m_n = m\cos\psi$
Bevel	$\tan\delta_1 = T_1/T_2$; $T_v = T/\cos\delta$
Worm	$VR = T_G/Z$; $\tan\lambda = L/\pi d_w$

9. Important Questions

- Explain gyroscopic couple; derive $C = I\omega\omega_p$. Explain the effect on aeroplane, ship (steering, pitching, rolling), two-wheeler and car taking a turn.
- State and prove the fundamental law of gearing. Compare involute and cycloidal profiles.
- Define path of contact, arc of contact and contact ratio ; derive expressions. What is interference? Derive the minimum number of teeth on a pinion.
- Describe helical, bevel, worm gears and rack & pinion with applications. Define virtual number of teeth.
- Explain simple, compound, reverted and epicyclic gear trains ; solve epicyclic problems by the tabular method.